

Aakash Sathish Shetty

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SUMMARY

Software Engineer with **3+ years of experience** building scalable backend services and APIs using **Python and SQL**, backed by strong computer science fundamentals (data structures, algorithms, OOP). Proven experience designing maintainable systems for concurrent traffic, deploying cloud services on AWS, and supporting production environments. Holds an **M.S. in Computer Science** with hands-on experience in **machine learning and NLP projects**.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C/C++, SQL, Perl

Frontend: React, HTML, CSS

Frameworks & Tools: Docker, Git, Django

Databases: PostgreSQL, MS SQL Server (MSSQL), MySQL

Backend: REST APIs, Authentication/Authorization, System Integration

Cloud & DevOps: AWS (EC2, S3), CI/CD pipelines

Testing: Unit Testing (JUnit, Cucumber)

Concepts: Data Structures, Algorithms, Code Reviews, Debugging, OOP, Scalability, Concurrency, API Design, Distributed Systems

EXPERIENCE

BusinessRate

May 2026 - Aug 2026

Software Engineer

- Migrated Doppler secrets to a standardized naming convention across 15 repositories and 4 environments, reducing configuration errors and easing onboarding for the team.
- Built a full-stack mail tracking integration using Lob's /events endpoint, combining scheduled backfill sync for historical data with real-time sync for ongoing events, and surfaced the analysis on an internal dashboard.
- Designed and implemented 8 new PostgreSQL tables across schemas to support the Lob integration, and altered multiple existing tables to resolve schema issues.
- Paired with another developer to build "Milestones," a cart-and-checkout flow extending an existing landing page, using Node.js, TypeScript, and Next.js.
- Stood up a new Stripe development environment with 25 test products replicated from production, configured across Doppler secrets and both database environments, enabling the team to safely test payment flows without using live products for the first time.
- Reviewed and merged 6-7 PRs per week from 2 teammates, helping the team ship changes faster and catch issues before production.

SowFin

Oct 2025 - May 2026

AI Software Engineer

- Architected a dual-provider external research pipeline for AI deal memos — Exa (parallel async fetching) — using Haiku for extraction and Sonnet for synthesis and reducing the processing time from 8 to 4 minutes (50% reduction).
- Optimized deal memo pipeline, hoisted permission checks to skip unnecessary LLM calls, pre-computed skip lists, and patched scores atomically in a single DB transaction and saving the cost by 85% for development.
- Designed and built **extensible Python backend APIs using reusable connector patterns** to integrate Google Drive, OneDrive, and Notion, enabling secure document syncing within the platform.
- Developed **React-based onboarding and account security flows**, including work-email verification, OTP authentication, and self-service password recovery.
- Collaborated with cross-functional stakeholders to communicate technical risks, trade-offs, and delivery timelines.
- Supported production deployments by **analyzing Docker logs**, diagnosing release issues, and shipping fixes in a fast-paced environment.
- Optimized Docker builds by introducing reusable base layers and dependency caching, **reducing build time from 40 minutes to 5 minutes (87% improvement)**; deployed services on **AWS EC2**.
- Participated in code reviews, debugged peers' code, and navigated a large codebase to deliver clean, maintainable changes while balancing speed and quality.
- Built a secure data room feature with access controls.

- Built a Stripe subscription system end-to-end — data models, webhook handler, subscription service, and React checkout flow; added promo code support with backend validation.
- Configured LLM prompt pipelines to generate targeted improvement recommendations from user document analyses, surfacing insights directly within the product UI.

Harman Connected Services

Dec 2020 – Dec 2023

Software Engineer

Full Time

- Built and maintained backend services handling **concurrent user requests** and **high-frequency database updates**.
- Provided production support by debugging live issues and fixing production bugs across a large shared codebase.
- Wrote unit tests and delivered constructive **code review feedback** on peers' pull requests.
- Automated deployments and data operations using **Python, shell scripting, and Docker**, improving deployment speed by **72%**.
- Optimized SQL queries and backend logic, improving data access efficiency by **18.7%**.
- Proactively documented systems, created onboarding materials, and shared knowledge to improve team execution and reduce ramp-up time.

EDUCATION

Arizona State University, Tempe, AZ

Jan 2024 - May 2025

Master of Science in Computer Science

Nitte Meenakshi Institute of Technology, Bangalore, KA, India

Aug 2016 - Sep 2020

Bachelor of Technology in Computer Science

PROJECTS

Reinforcement Learning Agent for a Pacman model

Mar 2024 - Apr 2024

- Implemented **REINFORCE** and **Actor-Critic** algorithms in Python.
- Experimented with reward design, training stability, and performance evaluation.

Customer Marketing Analysis

Jul 2024 - Aug 2024

- Analyzed the **UCI Adult dataset** using Pandas and NumPy.
- Performed uni- and multivariate analysis to study attribute relationships with salary.
- Built visualizations using **Matplotlib, Seaborn, and Plotly** to clearly communicate insights.

Optimal Joint Scheduling and Cloud Offloading

Aug 2019 - Mar 2020

- Designed a distributed mobile-cloud offloading approach, improving computational efficiency by **24.7%**.
- Published in **Springer ERCICA 2020**.

Mental Health Crisis Prediction

Feb 2025 - May 2025

- Built an end-to-end **machine learning pipeline** to classify mental health crisis signals from social media text.
- Compared **TF-IDF vs. embedding-based** features and benchmarked multiple supervised models.
- Analyzed feature importance and iterated preprocessing to improve model performance.

Parking App Prototype

Sep 2017 - Nov 2017

- Built a prototype **Android application** to allow users to search for and book parking spaces based on time requirements.
- Designed basic user flows for availability lookup, time-slot selection, and booking confirmation.